

Wang Jiefan

nafeij.me | wng.jiefan@gmail.com | linkedin.com/in/wang-jiefan | github.com/Nafeij | @wjiefan

EDUCATION

Bachelor of Computing

Honours w. Distinction (CGPA: 4.33) - Focus in Parallel Computing and Database Systems

National University of Singapore

Aug. 2021 – July 2025

EXTRA-CURRICULAR

Team Kent Ridge (NUS High-Performance Cluster Computing Interest Group)

June 2024 - July 2025

- Co-founder, captain of competition group of 20+ students.
- Wrote whitepapers on HPC technologies such as CUDA GPGPU programming, massively-distributed software parallelization frameworks such as OpenACC, advanced C++ features and memory-aware software optimizations.
- Designed and built private multi-node research cluster, administered with Ansible, running on Rocky Linux, serving a Gluster-distributed XFS filesystem and integrated with the Spack package manager, and monitored through Grafana.

ACHIEVEMENTS

ISC25 HPC-AI Online Student Cluster Competition

Aug. 2024 - June 2025

- 3rd Place.**
- Significantly rewrote the `μphys` module of ICON (the Icosahedral Nonhydrostatic Weather and Climate Model) using C++26 `std::par` compiler assisted offloading, paired with explicit memory-pinning CUDA routines to minimize inter-device buffer transfers.
- Achieved a roughly $\times 257$ mean runtime improvement ($>4\text{hrs} \rightarrow 56.1\text{s}$) when running the `atm_R2B08.nc` simulation using the DRKZ Levante system's GPU resources.

SC24 Indy-Student Cluster Competition

July - Nov. 2024

- Highest HPLinpack Score** (22.6 TFLOPS across 30 AMD EPYC-Milan CPU Nodes).
- Implemented custom MPI collective topologies to accelerate the HPL Linear algebra benchmark on the University of Indiana's JetStream2 supercomputer.
- Developed runtime architectures, kernel patches and parameter optimizations to enable scalable and efficient execution of the Nanoscale Molecular Dynamics simulation.

PROFESSIONAL EXPERIENCE

Backend Software Engineer - Global E-Commerce (Europe)

Singapore

TikTok Pte. Ltd

June 2025 - Present

- Implemented performance optimizations, such as asynchronous querying and task scheduling policies, to business logic of the TikTok shop seller platform that resulted in a $>60\%$ YoY increase in quarterly onboarding rate across all EU regions.
- Service OnCalls to maintain the stability and security of relevant systems, including performing data center disaster recovery and loss prevention/control.

Backend Platform Developer Intern

Singapore

Shopee Pte. Ltd.

May - Aug. 2024

- Developed Virtual Gateway feature for internal container networking platform in Go using gRPC and etcd, to automate and segregate the allocation of subnets to hundreds of in-production Virtual Private Clouds (VPCs) across global business units.
- Implemented Quality of Service (QoS) policies for internal container networking platform, by dynamically setting Differentiated Services Code Point values in IP headers using cgroup-attached eBPF programs, to enable client-side configuration of TCP traffic priorities for hundreds of baremetal VPC servers.

Teaching Assistant

Singapore

National University of Singapore

Aug. 2023 - May. 2025

- CS2109S Introduction to Machine Learning (AY23/24 Sem 1), CS3211 Parallel and Concurrent Programming (AY23/24, AY24/25 Sem 2), CS3210 Parallel Computing (AY24/25 Sem 1/2)
- Wrote new lesson material, created coding assignments, conducted weekly tutorial sessions and provided consultation to students on distributed systems programming topics such as cache-coherence protocols, the C++ memory model and lock-free programming.

PROJECTS

The Conq Programming Language | Rust, WebAssembly

Jan. – April 2024

- General-purpose programming language implemented with LLVM-MLIR toolchain and compiles to WASM with HTML Canvas bindings.

Order-Matching System | C++

Sep. – Nov. 2023

- Socket-based HFT-inspired multi-threading [order-matching engine](#) which utilizes custom extension of `std::priority_queue` that is thread-safe, supports arbitrary removal and iteration.

... and many more, at nafeij.me and github.com/Nafeij

TECHNICAL SKILLS

Languages: C/C++, Java, Kotlin, Go, Python, Typescript, Rust, PostgreSQL, SQLite, HTML\CSS, WebAssembly

Frameworks: React, OpenMP, OpenACC, React Native, Node.js, Next.js, gRPC, Flask, Ansible, Spack, Django, JUnit, NumPy, Pandas, PyTorch, Cilium

Technologies: Git, MPI, OpenMPI, eBPF, etcd, Kubernetes, ROCm, CUDA, Nvidia HPC SDK, Slurm, PBS, Singularity, Gluster, GraphQL, Jenkins, Firebase, OpenStack, JIRA

MISCELLANEOUS

I am a **Singaporean citizen**.